Elementary Problems
Triangle Geometry
Solutions

Version 1.0.1

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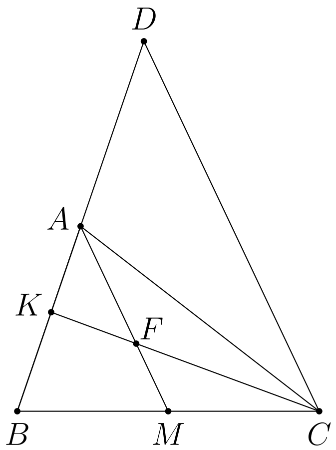
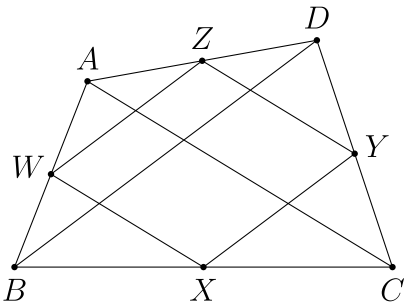
Basic Work

1. Let D be the point such that A is the midpoint of BD. By the midpoint theorem, we have $AM//DC$. It follows from $KA = KF$ that $AD = FC$. Hence, $AB = AD = FC$.



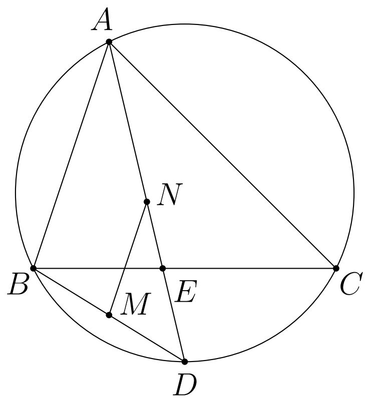
2. Since W and X are the midpoints of AB and BC respectively, we have $WX//AC$ by the midpoint theorem. Similarly, we have $ZY//AC$. Thus, $WX//ZY$. Similarly, we have $WZ//BD//XY$. These show $WXYZ$ is a parallelogram.



3. By the midpoint theorem, we have $AB // NM$. It follows that

$$\angle MNE = \angle BAD = \angle DAC = \angle DBC = \angle MBE.$$

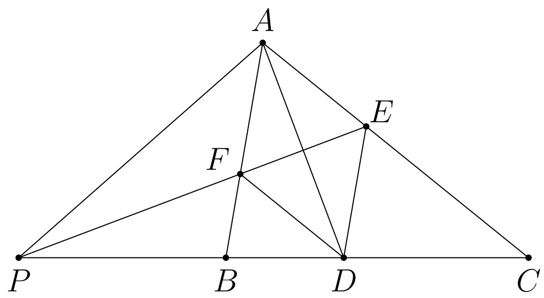
This shows B, M, E, N are concyclic.



4. Since $DE // BA$ and $DF // CA$, $AFDE$ is a parallelogram. Also, as AD bisects $\angle FAE$, $AFDE$ must be a rhombus. This shows EFP is the perpendicular bisector of AD. It follows from symmetry that

$$\angle PAB = \angle PAF = \angle FDP = \angle ACP = \angle ACB.$$

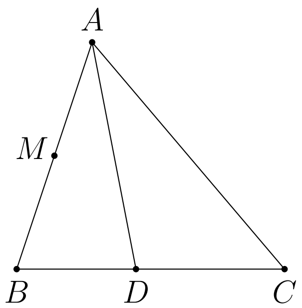
This shows PA is tangent to (ABC).



5. By the angle bisector theorem, we have $\dfrac{DB}{DC} = \dfrac{AB}{AC}$. Together with

$$DB + DC = BC = \frac{1}{2}(AB + AC),$$

we have $DB = \dfrac{1}{2}AB$ and $DC = \dfrac{1}{2}AC$. Hence, $DB = BM$.



6. Let D be the midpoint of AN. Then $AD = AK$. As $\angle KAD = 60°$, $\triangle AKD$ is equilateral. Hence, $DK = DA = DN$. This shows $\triangle AKN$ is right-angled with $\angle NKA = 90°$ and $\angle ANK = 30°$.

Next, we find that

$$\angle NMK = 180° - \angle KMB - \angle CMN = 180° - \angle MBK - \angle NCM = \angle BAC = 60°.$$

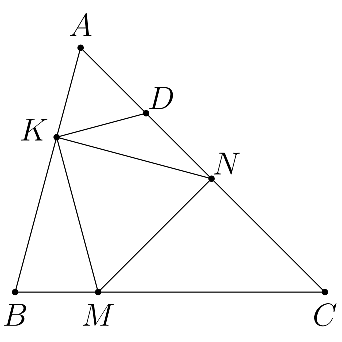
Together with $MK = MN$, we know that $\triangle KMN$ is equilateral.

Therefore, we have

$$\angle CBA = \frac{180° - \angle BKM}{2} = \frac{\angle MKN + \angle NKA}{2} = \frac{60° + 90°}{2} = 75°$$

and

$$\angle ACB = 180° - \angle BAC - \angle CBA = 180° - 60° - 75° = 45°.$$



7. Let D be the midpoint of BC. Since $DA = DB = DC$, point D is the centre of (ABC). As CD is tangent to (DNA), we have

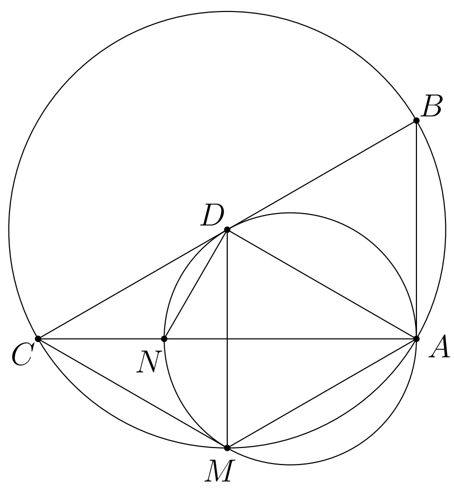
$$\angle CDN = \angle DAN = \angle ACD = \angle NCD.$$

This shows N lies on the perpendicular bisector of CD.

Next, as M lies on (ABC), we have $DM = DA$. Note that $\triangle ABD$ is equilateral as $DA = DB$ and $\angle DBA = 60°$. Since BD is tangent to (ADM), we find that $\angle AMD = \angle ADB = 60°$. This shows $\triangle ADM$ is equilateral.

From $DM = DC$ and $\angle CDM = \angle DAM = 60°$, $\triangle DCM$ is also equilateral. Thus, M lies on the perpendicular bisector of CD.

Therefore, MN is the perpendicular bisector of CD. In partiuclar, we have $MN \perp BC$.

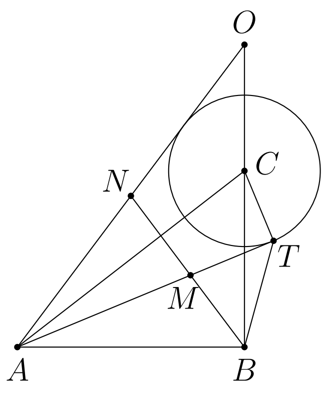


8. Let C be the centre of the circle and let N be the midpoint of AO.

Since $\angle CTA = 90° = \angle CBA$, points A, B, T, C are concyclic. Hence, we have

$$\angle MTB = \angle ATB = \angle ACB = \angle CAO + \angle AOB = \angle TAC + \angle AOB$$

since AC bisects the two tangents AT and AO.



As N is the midpoint of the hypotenuse of the right-angled $\triangle OAB$, we have $NB = NO$. Thus,

$$\angle TAC + \angle AOB = \angle TAC + \angle OBN = \angle TBC + \angle OBN = \angle TBM.$$

This yields $\angle MTB = \angle TBM$, so that $MB = MT$.

<u>Centroid</u>

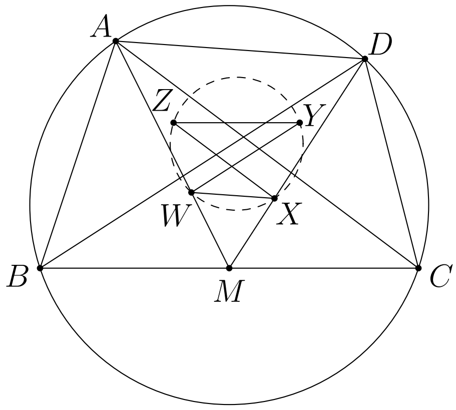
9. Let the centroids of $\triangle ABC$, $\triangle BCD$, $\triangle CDA$, $\triangle DAB$ be W, X, Y, Z respectively. Let M be the midpoint of BC. Note that W lies on AM and cuts the segment in the ratio $AW : WM = 2 : 1$. Similarly, X lies on DM and $DX : XM = 2 : 1$. From

$$\frac{AW}{WM} = 2 = \frac{DX}{XM},$$

we have $AD//WX$. Similarly, we have $BD//WY$, $CB//YZ$ and $ZX//AC$. Thus,

$$\angle ZXW = \angle CAD = \angle CBD = \angle ZYW.$$

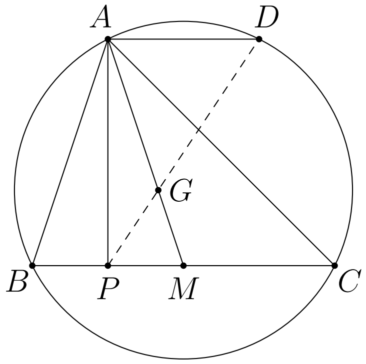
This show W, X, Y, Z are concyclic.



10. As $ABCD$ is cyclic and $AD//BC$, $ABCD$ is an isosceles trapezoid. Thus, the midpoint M of BC lies on the perpendicular bisector of AD. This implies $MP = \frac{1}{2}AD$. Hence,

$$\frac{MP}{AD} = \frac{1}{2} = \frac{MG}{AG}.$$

Together with $\angle GMP = \angle GAD$, we get $\triangle GMP \sim \triangle GAD$. From this, we obtain $\angle PGM = \angle DGA$. Therefore, P, G, D are collinear.

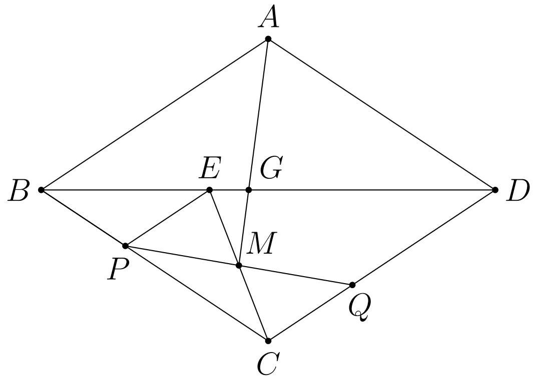


11. Let E be the point on BD such that $PB = PE$. Then

$$\angle BEP = \angle PBE = \angle BDC.$$

This implies $PE//CQ$, and hence $EPCQ$ is a parallelogram since $PE = CQ$. Thus, the midpoint M of PQ is the midpoint of CE.

Let G be the centroid of $\triangle APQ$. Then G lies on AM with $AG : GM = 2 : 1$. Hence, G is also the centroid of $\triangle AEC$. Since the midpoint of AC lies on BD, G lies on BD.



12. Let D and E be the midpoints of AB and AC respectively. Let G be the centroid of $\triangle ABC$. As $\dfrac{BM}{MA} < 1$, point M lies between B and D. Similarly, N lies between C and E. Let $b = AC$ and $c = AB$. We have

$$\frac{EN}{NA} \times \frac{AM}{MB} \times \frac{BG}{GE} = \frac{\frac{b}{2} - CN}{b - CN} \times \frac{c - BM}{BM} \times 2$$

as G cuts the segment BE in the ratio $2 : 1$. The given condition is the same as

$$\frac{BM}{c - BM} + \frac{CN}{b - CN} = 1.$$

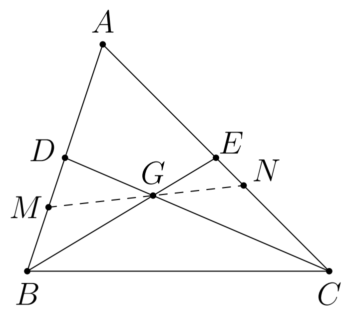
This can be rewritten as

$$\frac{BM}{c - BM} = \frac{b - 2CN}{b - CN}.$$

From this, we find that

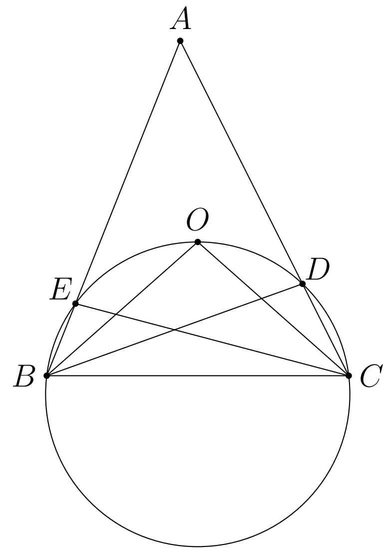
$$\frac{EN}{NA} \times \frac{AM}{MB} \times \frac{BG}{GE} = \frac{\frac{b}{2} - CN}{b - CN} \times \frac{b - CN}{b - 2CN} \times 2 = 1.$$

From the configuration, we can apply Menelaus' theorem. This shows M, G, N are collinear.



<u>Circumcentre</u>

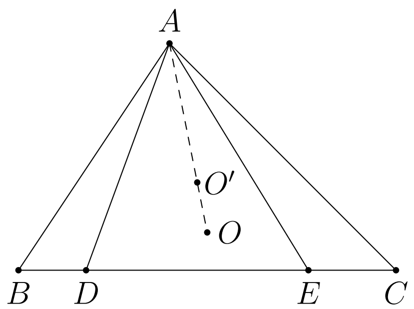
13. Since $DA = DB$, we have $\angle BDC = \angle DBA + A = 2A$. Similarly, we have $\angle BEC = 2A$. Together with $\angle BOC = 2A$, points E, B, C, D, O are concyclic.



14. We have

$$\begin{aligned}\angle DAO &= \angle BAO - \angle BAD \\ &= (90^\circ - \angle ACB) - \angle EAC \\ &= 90^\circ - \angle AED = \angle DAO'.\end{aligned}$$

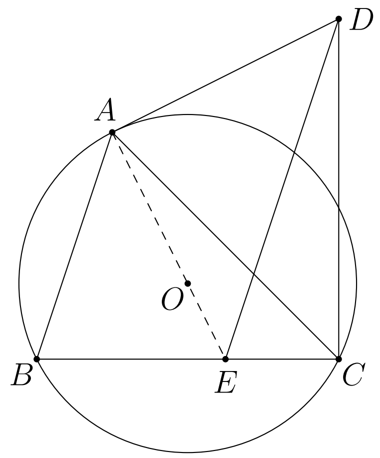
This implies $\angle DAO' = \angle DAO$, i.e. A, O, O' are collinear.



15. Since

$$\angle CAD = \angle CBA = \angle CED,$$

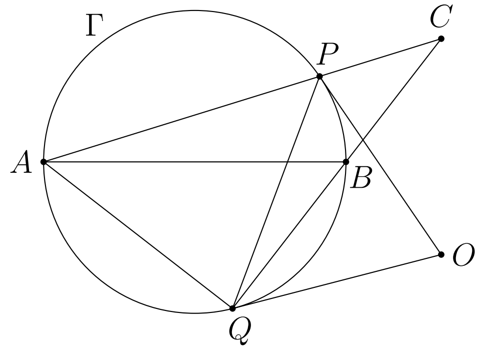
points A, E, C, D are concyclic. This yields $\angle EAD = 180^\circ - \angle DCE = 90^\circ$. As $\angle OAD = 90^\circ$ from the tangent, points A, O, E are collinear.



16. As O is the circumcentre of $\triangle CPQ$, we have

$$\angle QPO = \frac{180^\circ - \angle POQ}{2} = 90^\circ - \angle PCQ.$$

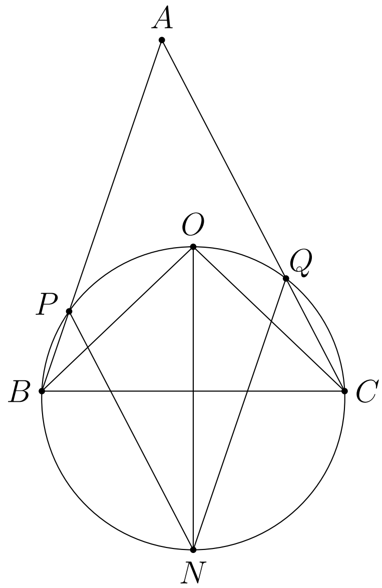
This is equal to $\angle QAP$ since $\angle BQA = 90^\circ$ from the fact that AB is a diameter of Γ. Therefore, we have $\angle QPO = \angle QAP$, which shows that OP is tangent to Γ. As $OQ = OP$, the line OQ is also tangent to Γ.



17. Since O lies on the perpendicular bisector of BC, so is N. Therefore, we have

$$\angle NPA = 180^\circ - \angle BPN = 180^\circ - \angle BON = 180^\circ - \frac{1}{2}\angle BOC = 180^\circ - A.$$

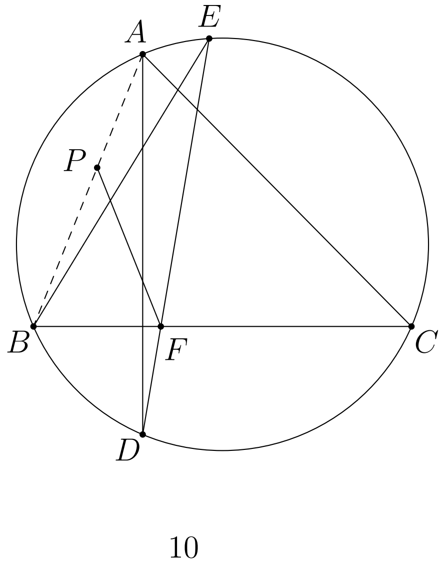
This implies $NP//QA$. Similarly, $NQ//PA$. Hence, $APNQ$ is a parallelogram.



18. Let P be the circumcentre of $\triangle BFE$. Then we have

$$\angle FBP = \frac{180^\circ - \angle BPF}{2} = 90^\circ - \angle BEF = 90^\circ - \angle BAD = \angle CBA.$$

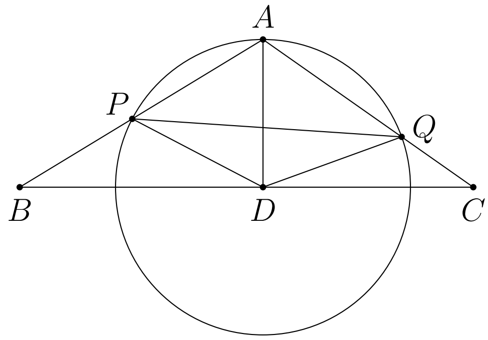
This shows A, P, B are collinear, i.e. P lies on AB.



19. Since D is the centre of (APQ), we have

$$\angle AQP = \frac{1}{2}\angle ADP = \frac{1}{2}(180^\circ - 2\angle PAD) = 90^\circ - \angle PAD = \angle CBP.$$

Together with the common angle $\angle PAQ = \angle CAB$, we get $AQP \sim \triangle ABC$.



20. Firstly, we have

$$\angle FCD = \angle BCD = \angle BAD = 90^\circ - B = \angle OAC = \angle EAC.$$

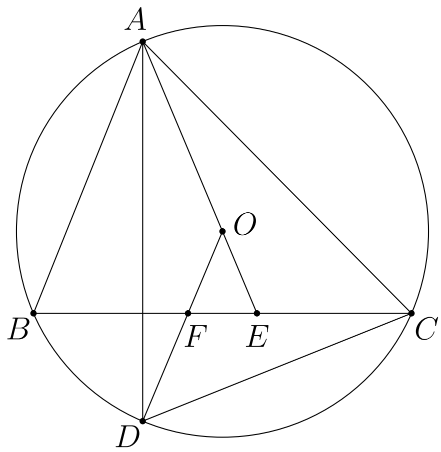
Secondly, we have

$$\angle CDF = \angle CDA - \angle ODA = \angle CBA - \angle DAO$$

since $OA = OD$. This yields

$$\angle CDF = B - (\angle BAO - \angle BAD) = B - (90^\circ - C) + (90^\circ - B) = C = \angle ACE.$$

Hence, $\triangle CFD \sim \triangle AEC$.



21. WLOG assume $\angle ADB \leqslant 90°$. Then

$$\angle BAX = 90° - \angle ADB$$

and

$$\angle CAY = 90° - \frac{1}{2}\angle AYC = 90° - \frac{1}{2}(360° - 2\angle CDA) = \angle CDA - 90°.$$

Therefore, we have

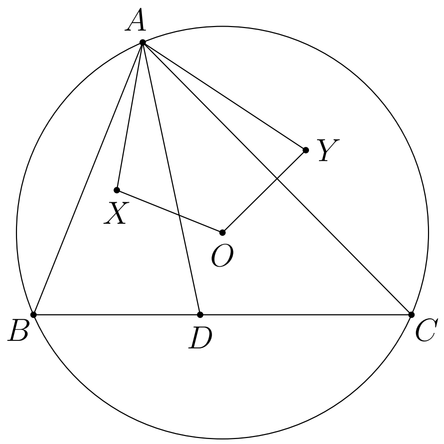
$$\angle XAY = A - \angle BAX + \angle CAY = A - (90° - \angle ADB) + (\angle CDA - 90°) = A$$

since $\angle ADB + \angle CDA = 180°$.

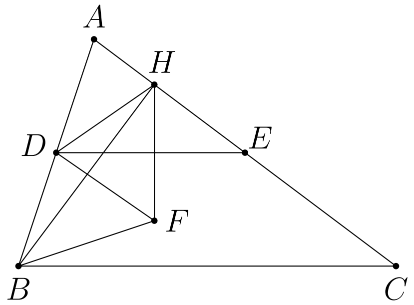
As O and X lie on the perpendicular bisector of AB, we have $OX \perp AB$. Similarly, we have $OY \perp AC$. Hence,

$$\angle YOX = 360 - 90° - 90° - A = 180° - A = 180° - \angle XAY.$$

Thus, A, X, O, Y are concyclic.



22. Note that $DA = DB = DH$ as D is the midpoint of the hypotenuse of the right-angled $\triangle AHB$. Since $DH = DF$, points A, B, F, H are concyclic. Indeed, they lie on a circle with centre D.



Thus, we have

$$\angle FBA = \angle FHE = 90^\circ - \angle AED = 90^\circ - C$$

since $DE//BC$ by the midpoint theorem. It follows that BF passes through the circumcentre O of $\triangle ABC$ (by noting that $\angle OBA = 90^\circ - C$).

23. Let the reflections of E in AB, BC, CD, DA be E_1, E_2, E_3, E_4 respectively.

Since $AE_1 = AE = AE_4$, A is the circumcentre of $\triangle E_1EE_4$. This implies

$$\angle E_1E_4E = \frac{1}{2}\angle E_1AE = \angle BAE$$

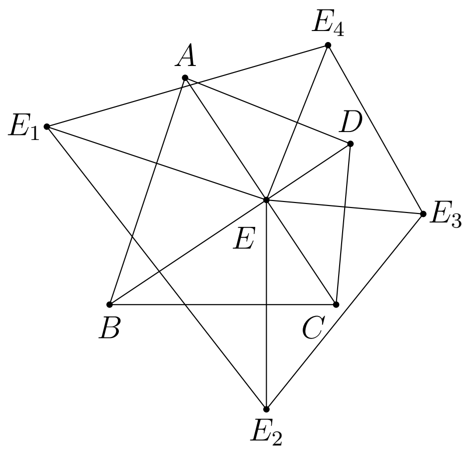
due to the reflection. Similarly, we have

$$\begin{aligned}\angle EE_4E_3 &= \angle EDC,\\ \angle EE_2E_1 &= \angle EBA,\\ \angle E_3E_2E &= \angle DCE.\end{aligned}$$

Hence,

$$\begin{aligned}\angle E_1E_4E_3 + \angle E_3E_2E_1 &= \angle E_1E_4E + \angle EE_4E_3 + \angle EE_2E_1 + \angle E_3E_2E\\ &= \angle BAE + \angle EDC + \angle EBA + \angle DCE\\ &= (\angle BAE + \angle EBA) + (\angle EDC + \angle DCE)\\ &= 90^\circ + 90^\circ = 180^\circ.\end{aligned}$$

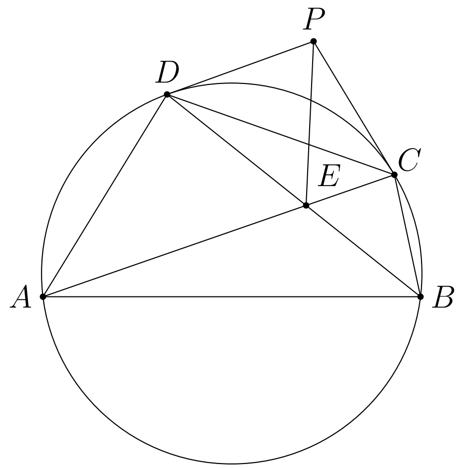
This shows E_1, E_2, E_3, E_4 are concyclic.



24. We have

$$\begin{aligned}2\angle CED &= (\angle EAD + \angle ADE) + (\angle CBE + \angle ECB)\\ &= \angle PCD + 90^\circ + \angle CDP + 90^\circ\\ &= 180^\circ + (180^\circ - \angle DPC)\\ &= 360^\circ - \angle DPC.\end{aligned}$$

So the reflex $\angle DPC$ is twice of $\angle CED$. As $PC = PD$, this implies P is the circumcentre of $\triangle CDE$. Hence, $PC = PE$.



25. Firstly, from

$$\angle ONB = \angle OAC = 90^\circ - B,$$

we find that $NO \perp AB$. Since O lies on the perpendicular bisector of AB, the same holds for the point N.

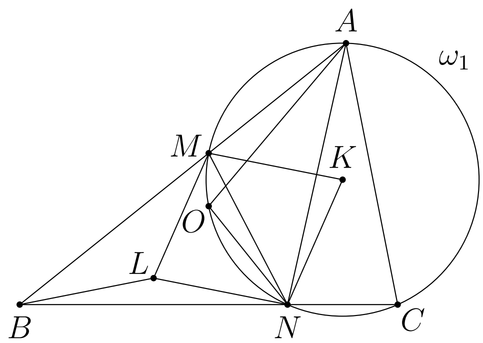
Next, we have

$$\angle NLM = \angle MKN = 2\angle MAN = 2B$$

as $NA = NB$ from above. Since we also have $LM = LN$, point L is the circumcentre of $\triangle BNM$. Hence,

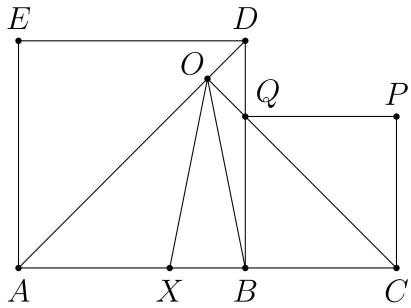
$$\angle NBL = 90^\circ - \angle BMN = 90^\circ - C$$

since M, N, C, A are concyclic. This implies $BL \perp AC$.



26. Since the perpendicular bisector of EB is AD, and the perpendicular bisector of PB is CQ, the circumcentre O of $\triangle EBP$ is the intersection of AD and CQ.

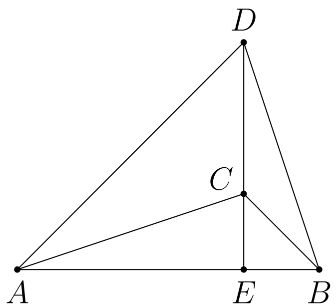
As $\angle OCA = \angle CAO = 45°$, O lies on the perpendicular bisector of AC. From the condition $AX = BC$, O also lies on the perpendicular bisector of XB. This yields $OX = OB$. Hence, X lies on (EBP). In other words, E, X, B, P are concyclic.



Orthocentre and Nine Point Circle

27. Since $\angle ADC + \angle BAD = 90°$, we have $CD \perp AB$. Similarly, we have $BC \perp AD$. This shows C is the orthocentre of $\triangle DAB$.

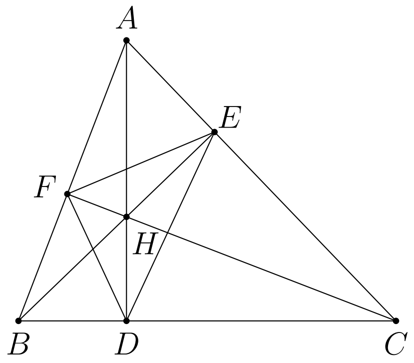
Let CD meet AB at E. Then $\triangle AEC \sim \triangle DEB$. In addition, from $\angle CBE = 45°$ and $\angle BEC = 90°$, we have $EC = EB$. Hence, $\triangle AEC \cong \triangle DEB$. Thus, $AC = DB$.



28. Note that B, D, H, F; C, E, H, D; and B, C, E, F are groups of concyclic points respectively. Thus, we have

$$\angle EDH = \angle ECH = \angle HBF = \angle HDF.$$

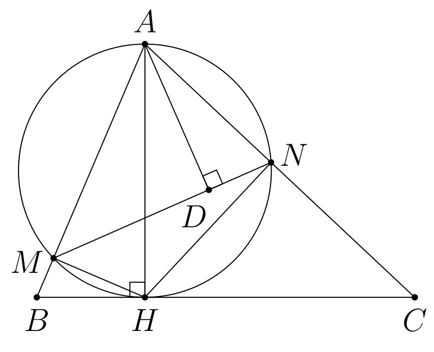
This implies DH bisects $\angle EDF$. Similarly, EH bisects $\angle FED$. Hence, H is the incentre of $\triangle DEF$.



29. Let L_A meet MN at D. Since AH is a diameter of (AMH), we have $\angle AMH = 90°$, which is equal to $\angle ADN$. From the concyclic points, we also have $\angle AHM = \angle AND$. These show $\triangle AMH \sim \triangle ADN$. Thus,

$$\angle DAN = \angle MAH = 90° - B.$$

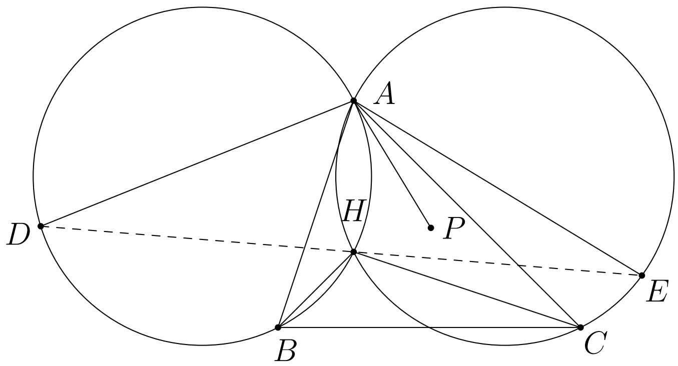
It follows that L_A passes through the circumcentre O of $\triangle ABC$. Hence, L_A, L_B, L_C are concurrent at O.



30. We have

$$\begin{aligned}\angle DHE &= \angle DHB + \angle BHC + \angle CHE = \angle DAB + \angle BHC + \angle CAE \\ &= \angle BAP + \angle BHC + \angle PAC = \angle BAC + \angle BHC = 180^\circ.\end{aligned}$$

Hence, D, H, E are collinear.



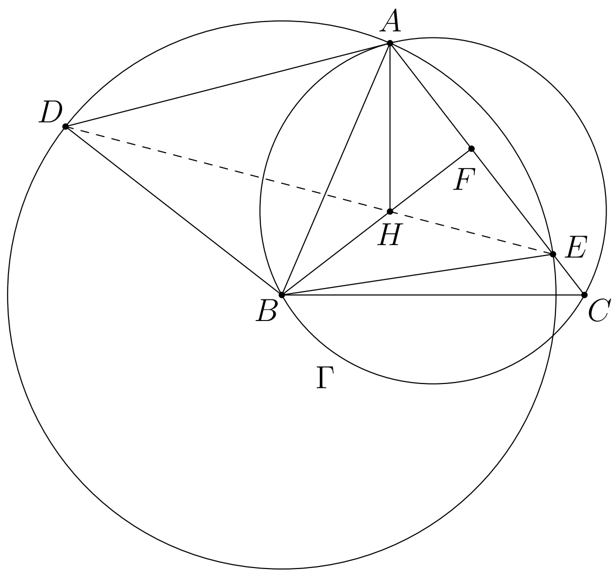
31. Since $BA = BE$ and $BH \perp AE$, BH is the perpendicular bisector of AE. Therefore, we have

$$\angle AEH = \angle HAE = 90^\circ - C.$$

Next, as B is the centre of (ADE), we have

$$\angle AED = \frac{1}{2}\angle ABD = \frac{1}{2}(180^\circ - 2\angle DAB) = 90^\circ - \angle DAB = 90^\circ - C$$

as DA is tangent to Γ. Hence, we obtain $\angle AEH = \angle AED$. Thus D, H, E are collinear.



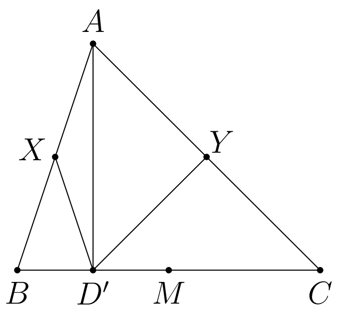
32. Let D' be the foot of the altitude from A. As $\angle AD'B = 90°$ and $XA = XB$, we have $XD' = XA$. Thus, $\angle AD'X = \angle XAD'$. Similarly, we have $\angle YD'A = \angle D'AY$. This implies

$$\angle YD'X = \angle YD'A + \angle AD'X = \angle D'AY + \angle XAD' = \angle BAC.$$

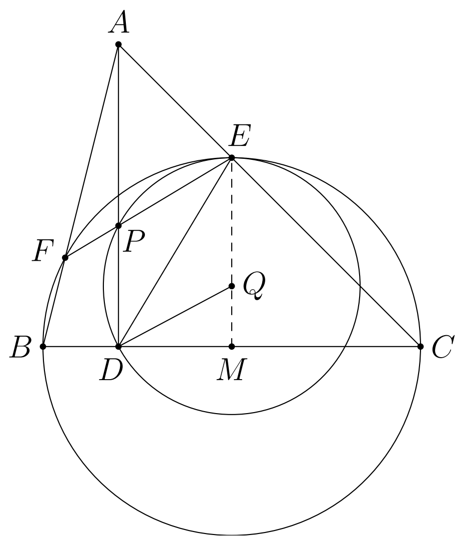
Let M be the midpoint of BC. Then $\triangle MXY \cong \triangle AYX$. From this, we obtain $\angle YMX = \angle BAC$.

Any point D satisfying $\angle YDX = \angle BAC$ must lie on the arc XMY of (XMY). Since (XMY) and BC have at most two intersection points, there are only two possible positions of D. From above, D can only be D' or M. The latter case is rejected from the given constraint. Thus, $D' = D$ and $AD \perp BC$.

(Indeed, for the limiting case $AB = AC$, the points D' and M coincide. However, it is obvious by symmetry that there can be at most one point D lying on BC satisfying $\angle YDX = \angle BAC$. So this case is rejected.)



33. Let Q be the circumcentre of $\triangle PDE$, and let M be the midpoint of BC. Recall that M is the centre of $(BCEF)$. Thus, the conclusion holds if and only if E, Q, M are collinear.



Note that we have

$$\begin{aligned}\angle DEQ &= \frac{180^\circ - \angle EQD}{2} = \frac{180^\circ - (360^\circ - 2\angle DPE)}{2} \\ &= \angle DPE - 90^\circ = \angle PAE + \angle AEF - 90^\circ \\ &= (90^\circ - C) + B - 90^\circ = B - C\end{aligned}$$

as B, C, E, F are concyclic. Also, we have

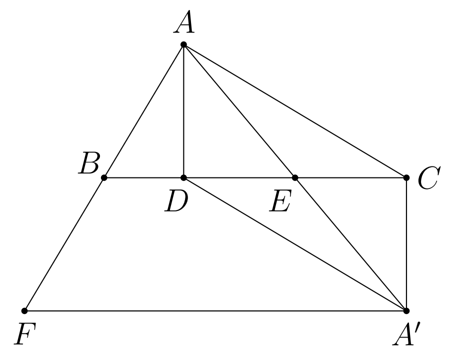
$$\angle DEM = \angle DEC - \angle MEC = B - C$$

since A, B, D, E are concyclic and $MC = ME$. These show $\angle DEQ = \angle DEM$. Thus, E, Q, M are collinear. The result then follows.

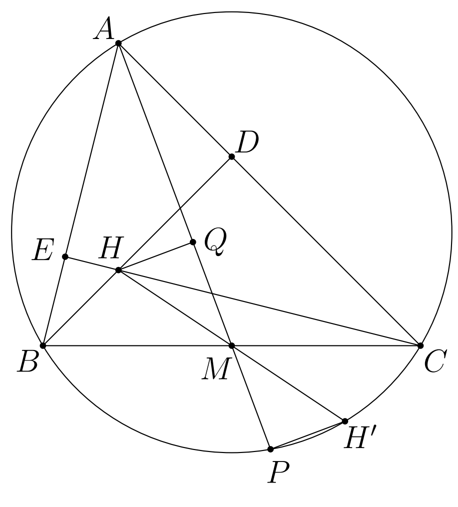
34. Let A' be the reflection of A in the point E. Since E is the midpoint of CD and AA', $ADA'C$ is a parallelogram. This yields $A'D // CA$ and hence $A'D \perp AF$.

    Also, by the midpoint theorem, we have $BE // FA'$. This implies $AD \perp FA'$.

    Therefore, D is the orthocentre of $\triangle AFA'$. It follows that $FD \perp AE$.



35. Let H' be the reflection of H in the point M. Recall that H' lies on (ABC) and AH' is a diameter of the circumcircle. Thus, we have $AP \perp PH'$.



As M is the midpoints of both PQ and HH', $QHPH'$ is a parallelogram. In particular, we have $HQ//PH'$. Hence, $HQ \perp AP$.

From $\angle AQH = \angle ADH = \angle AEH = 90°$, points A, E, H, Q, D are concyclic.

36. Let X be the foot of the altitude from A to BC. Let Y be the foot of the altitude from A to DE. Firstly, from

$$\angle EDA = \angle PDC = 90° - C = \angle CBH$$

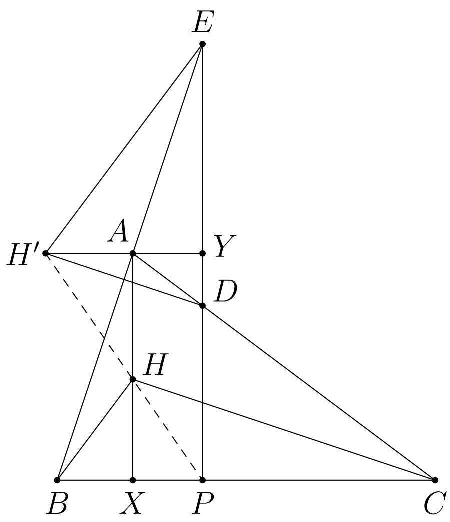
and

$$\angle DAE = 180° - A = \angle BHC,$$

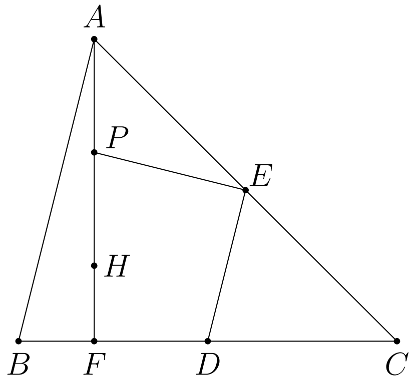
we have $\triangle ADE \sim \triangle HBC$. Since H' is the orthocentre of $\triangle ADE$ and A is the orthocentre of $\triangle HBC$, the figures $H'ADYE$ and $AHBXC$ are similar. Hence,

$$\frac{AH}{HX} = \frac{H'A}{AY} = \frac{H'A}{XP}$$

since $AXPY$ is a rectangle. Together with $\angle H'AH = 90° = \angle PXH$, we obtain $\triangle HAH' \sim \triangle HXP$. Thus, $\angle AHH' = \angle XHP$ and hence H', H, P are collinear.



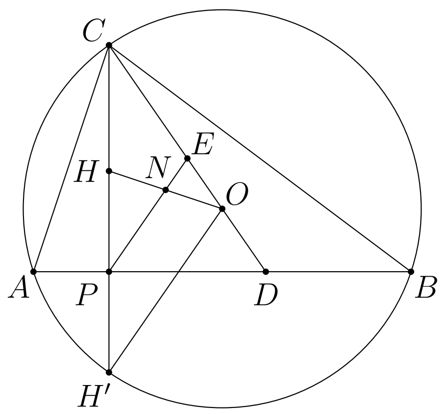
37. Let F be the foot of the altitude from A. Recall, that P, F, D, E lie on the nine point circle of $\triangle ABC$. As $\angle DFP = 90°$, we have $\angle PED = 180° - \angle DFP = 90°$.



38. Let H' be the reflection of H in the line AB. Recall that H' lies on (ABC). Since $\angle DPC = 90°$ and $CE = ED$, we have $CE = PE$. It is obvious that $CO = H'O$ as they are radii of the circumcircle. Therefore, we have

$$\angle EPC = \angle PCE = \angle H'CO = \angle OH'C.$$

This implies $PE//H'O$. Let N be the intersection of PE and HO. As $PN//H'O$ and $HP = PH'$, we have $HN = NO$. This shows EP bisects OH.

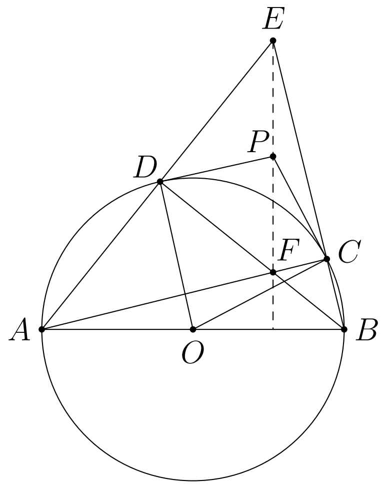


39. Note that there are several possible configurations. We only give a proof for the configuration as shown. The other cases can be proved in the same way.

Let O be the centre of the circle. Let the extension of AD and BC meet at E. Let F be the intersection of AC and BD, and let P be the intersection of the tangents at C and D.

Since AB is a diameter, we have $BD \perp AE$ and $AC \perp BE$. Thus, C and D are the feet of the altitudes of $\triangle EAB$, and F is its orthocentre. As O is the midpoint of AB, (DOC) is the nine point circle of $\triangle EAB$.

Next, we have $\angle PDO = \angle PCO = 90°$ from the tangents. This shows P lies on (DOC), and PO is a diameter of the nine point circle. Thus, P is the midpoint of EF. Since F is the orthocentre of $\triangle EAB$, we have $PF \perp AB$.

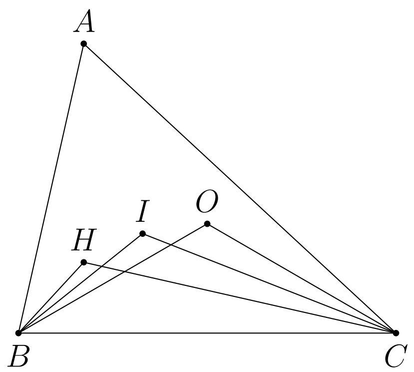


Incentre and Excentres

40. We have

$$\angle BHC = 180° - A = 120°,$$
$$\angle BIC = 90° + \frac{1}{2}A = 120°,$$
$$\angle BOC = 2A = 120°.$$

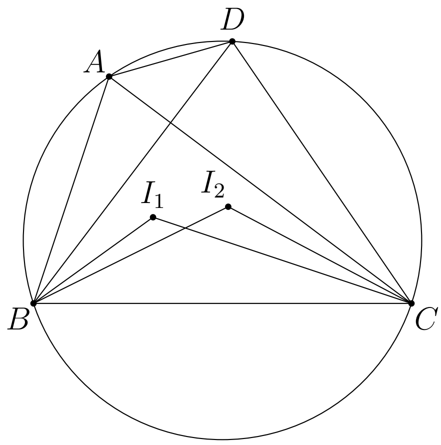
Therefore, B, C, O, I, H are concyclic.



41. We have

$$\angle BI_1C = 90^\circ + \frac{1}{2}\angle BAC = 90^\circ + \frac{1}{2}\angle BDC = \angle BI_2C.$$

This implies B, C, I_2, I_1 are concyclic.



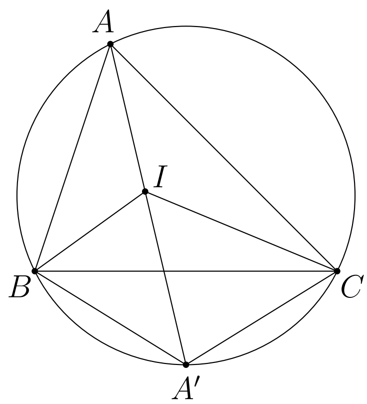
42. It is well-known that A' lies on the circumcircle of $\triangle ABC$. Indeed, as

$$\angle BIC = 90^\circ + \frac{A}{2}$$

is obtuse, we have

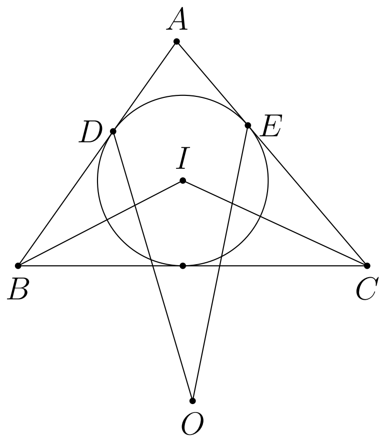
$$\angle CA'B = 360^\circ - 2\angle BIC = 180^\circ - A.$$

This implies A, B, A', C are concyclic. Similarly, B' and C' lie on the circumcircle of $\triangle ABC$. The result is now obvious.



43. Recall that O lies on the internal angle bisector of $\angle BAC$. By $\angle OAD = \angle OAE$, $AD = AE$ and the common side AO, we have $\triangle OAD \cong \triangle OAE$. Therefore, we find that $\angle ODA = \angle OEA$. Thus,

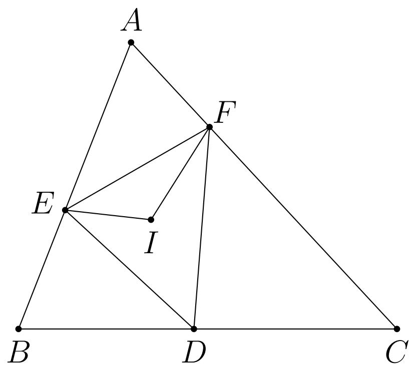
$$\angle ODB = 180^\circ - \angle ODA = 180^\circ - \angle OEA = \angle OEC.$$



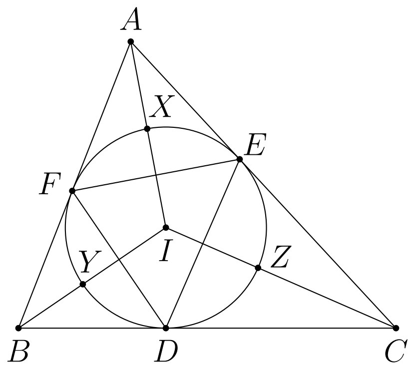
44. We have

$$\begin{aligned}\angle FIE &= 90^\circ + \frac{1}{2}\angle FDE \\ &= 90^\circ + \frac{1}{2}(180^\circ - \angle EDB - \angle CDF) \\ &= 180^\circ - \frac{1}{2}(180^\circ - 2B) - \frac{1}{2}(180^\circ - 2C) \\ &= B + C = 180^\circ - \angle EAF\end{aligned}$$

by using $DB = DE$ and $DC = DF$. Hence, A, E, I, F are concyclic.



45. Since AI is the perpendicular bisector of EF, point X is the midpoint of the arc EF of Γ. This implies DX bisects $\angle EDF$. Similarly, EY and FZ are the internal angle bisectors of $\angle FED$ and $\angle DFE$ respectively. Hence, DX, EY, FZ are concurrent at the incentre of $\triangle DEF$.



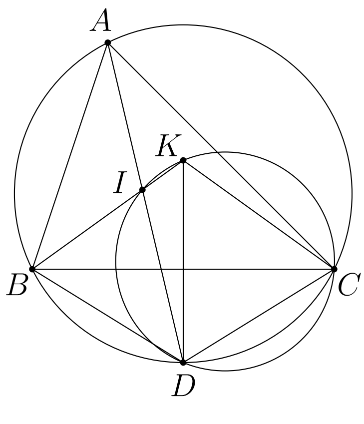
46. Recall that $DB = DI = DC$. We have

$$\angle KCD = \angle BID = \angle IBD = \angle KBD$$

since $DB = DI$. Also, we have $\angle DCB = \angle DBC$ from $DB = DC$. Hence,

$$\angle BCK = \angle DCK - \angle DCB = \angle DBK - \angle DBC = \angle CBK.$$

It follows that $BK = CK$.



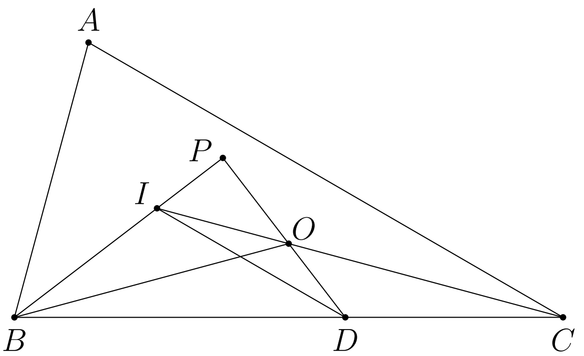
47. Note that C, O, I are collinear since $AC = BC$. Let BI meet DO at P. We have

$$\begin{aligned}\angle IOD &= \angle IPO + \angle OIP = 90^\circ + (180^\circ - \angle BIC)\\ &= 270^\circ - \left(90^\circ + \frac{A}{2}\right)\\ &= 180^\circ - \frac{B}{2} = 180^\circ - \angle DBI.\end{aligned}$$

This shows I, B, D, O are concyclic. It follows that

$$\angle IDB = \angle IOB = \angle OCB + \angle CBO = 2\angle OCB = \angle ACB.$$

Thus, $ID // AC$.

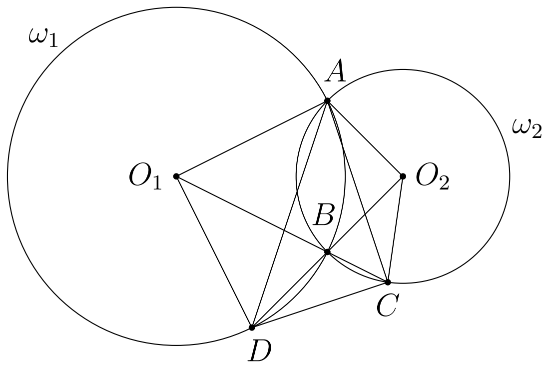


48. Due to symmetry, we have $\angle O_1AO_2 = \angle O_2BO_1$. Hence,

$$\angle O_1AO_2 = 180^\circ - \angle CBO_2 = 180^\circ - \angle O_2CB.$$

This implies A, O_1, C, O_2 are concyclic. Similarly, A, O_1, D, C, O_2 are concyclic.

As A, D, C, O_2 are concyclic and $O_2A = O_2C$, the line DO_2 bisects $\angle CDA$. Similarly, CO_1 bisects $\angle ACD$. Hence, B is the incentre of $\triangle ADC$.



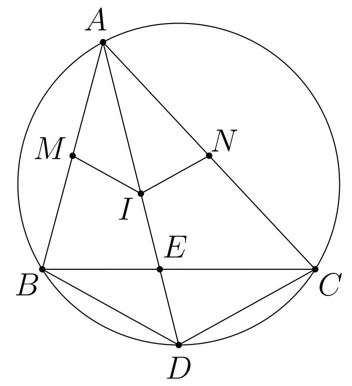
49. Let AI meet BC at E, and let AI meet (ABC) at D. By the angle bisector theorem, we have $\dfrac{EB}{EC} = \dfrac{AB}{AC}$. Together with $EB + EC = BC = \dfrac{1}{2}(AB + AC)$, we have $EB = \dfrac{1}{2}AB$ and $EC = \dfrac{1}{2}AC$.

From $\angle BAE = \dfrac{A}{2} = \angle DAC$ and $\angle EBA = \angle CBA = \angle CDA$, the triangles ABE and ADC are similar. This yields $\dfrac{AD}{DC} = \dfrac{AB}{BE} = 2$. Recall that $DC = DI$. Therefore, we obtain

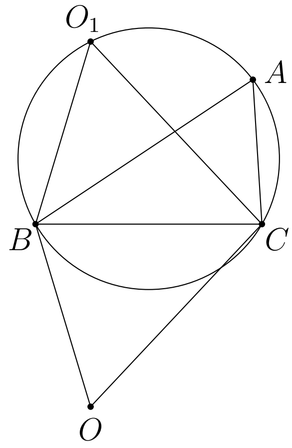
$$DI = DC = \frac{1}{2}AD.$$

In other words, I is the midpoint of AD.

Now, the homothety with centre A and ratio $\dfrac{1}{2}$ maps points B, D, C to points M, I, N respectively. As A, B, D, C are concyclic, A, M, I, N are also concyclic.



50. As O is the A-excentre, we have $\angle COB = 90^\circ - \dfrac{A}{2}$.



Hence,

$$A = \angle BO_1C = \angle COB = 90^\circ - \frac{A}{2}.$$

Solving this, we obtain $A = 60^\circ$.

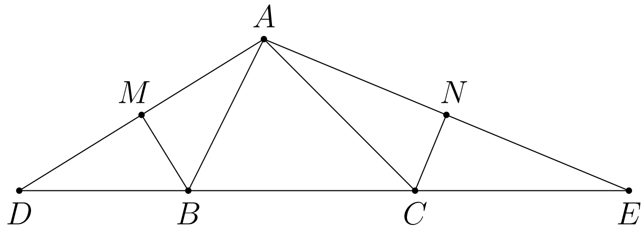
51. Let the extension of AM and BC meet at D, and let the extension of AN and BC meet at E.

    As BM bisects $\angle ABD$ and $BM \perp AD$, we have $BD = BA$. Similarly, $CE = CA$. This shows

    $$DE = DB + BC + CE = AB + BC + CA$$

    is the perimeter of $\triangle ABC$.

    Note that M and N are the midpoints of AD and AE respectively. By the midpoint theorem, we have $MN = \frac{1}{2}DE$, which is the semiperimeter of $\triangle ABC$.



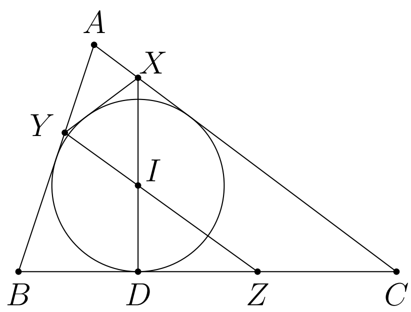
52. Note that I is the A-excentre of $\triangle AYX$. This yields

    $$\angle XIY = 90^\circ - \frac{\angle YAX}{2}.$$

    Hence, we have

    $$\angle IZB = 90^\circ - \angle DIZ = 90^\circ - \angle XIY = \frac{A}{2} = \angle IAB.$$

    Together with $\angle ZBI = \angle ABI$ and $BI = BI$, we have $\triangle IBZ \cong \triangle IBA$. Therefore, $BZ = BA$.



53. Firstly, as AE bisects $\angle CAD$, we have

$$\angle CAE = \frac{1}{2}\angle CAD = \frac{1}{2}\angle ACB.$$

As CI bisects $\angle ACB$, we have

$$\frac{1}{2}\angle ACB = \angle ACI = \angle ICB.$$

From $\angle CAE = \angle ACI$, we have $AE//IC$. From $\angle CAE = \angle ICB = \angle CIE$, points A, I, C, E are concyclic. Thus, $AICE$ is an isosceles trapezoid.

Next, since E is the A-excentre of $\triangle ACD$, we have

$$\angle AEC = \frac{1}{2}\angle ADC.$$

From above, $AICE$ is an isosceles trapezoid. Thus, $\angle IAE = \angle AEC$. As

$$\angle IAE = \angle IAC + \angle CAE = \frac{1}{2}\angle BAC + \frac{1}{2}\angle CAD = \frac{1}{2}\angle BAD,$$

we obtain $\angle ADC = \angle BAD$. Since $AD//BC$, this shows $ABCD$ is an isosceles trapezoid.